

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : ESEE101 Basic Electrical Engineering

UPID : 001009

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) What is fuse?
- (II) What is the efficiency of power transfer when maximum transfer of power occurs?
- (III) In a series R-L-C circuit, what is the condition for resonance?
- (IV) What is core type transformer?
- (V) What is basic principle of operation of Alternators / DC Generators?
- (VI) What is meant by duty-cycle?
- (VII) What is the function of a transformer?
- (VIII) Define synchronous speed.
- (IX) What are the main classification of inverter?
- (X) What is secondary cell?
- (XI) What will be the impedance of a coil containing resistance of 8.0Ω and an Inductive reactance of 6.0Ω ?
- (XII) The full-load copper loss of a transformer is 1600 W. At half-load, what will be the copper loss?

Group-B (Short Answer Type Question)*Answer any three of the following*

[5 x 3 = 15]

2. What is the significance of form factor in ac circuit? [5]
3. What are the salient features of distribution transformer? [5]
4. An AC current is expressed as $i=14.14\sin 314t$. Determine its a) RMS value, b) frequency, c) instantaneous value at $t=2$ ms. [5]
5. What is meant by Regulation in a Transformer? [5]
6. Why are iron losses considered as constant losses in transformer? [5]

Group-C (Long Answer Type Question)*Answer any three of the following*

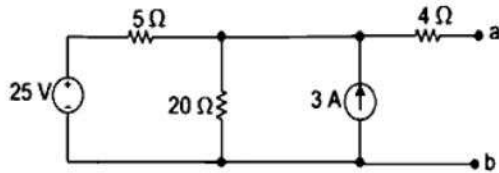
[15 x 3 = 45]

7. (a) Explain the principle of series resonance. [5]
(b) What is half power frequency? [4]
(c) A coil of inductance 9 H and resistance 50Ω in series with a capacitor is supplied at constant voltage from a variable frequency source. If the maximum current of 1A occurs at 75 Hz, find the frequency when the current is 0.5 A. [6]
8. (a) What is shell type transformer? Explain in detail. [6]
(b) The primary and secondary of a 25 kVA transformer has 500 and 40 turns, respectively. If the primary is connected to 3000 V, 50 Hz mains, calculate (i) Primary and secondary currents at full load; (ii) The secondary emf and (iii) The maximum flux in the core. Neglect magnetic leakage, resistance of the winding and the primary no-load current in relation to the full load current. [9]
9. (a) Justify that the average active power consumed by a purely inductive circuit is zero. [7]
(b) Explain the phasor diagram of RC series circuit with a neat circuit diagram. [8]

10. (a) A 100 kVA, 3300/200 volt, 50 Hz single phase transformer has 40 turns on the secondary, calculate: [9]
- (i) the values of primary and secondary currents.
 - (ii) the number of primary turns,
 - (iii) the maximum value of the flux.
- If the transformer is to be used on a 25 Hz system, calculate
- (iv) the primary voltage, assuming that the flux is increased by 10%,
 - (v) the kVA rating of the transformer assuming the current density in the windings to be unaltered.

(b) Derive the emf equation of a single phase transformer. [6]

11. (a) Derive the Norton equivalent circuit of the given figure. [8]



(b) Find Thevenin equivalent circuit for the following network. [7]

